Abhinav Raundhal

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Education

International Institute of Information Technology, Hyderabad — Hyderabad, India
B.Tech. in Computer Science and Engineering (Honours), CGPA: 9.64/10.0 — *Oct. 2022 – Jun. 2026*

- **Best All-Rounder Award** (Institute Gold Medal) for excellence in academics and extracurricular activities.
- **Dean's List** in all 8 semesters (DL1: 4, DL2: 3, Merit List: 1). Perfect 10 SGPA in 2 semesters.

Publications

STRinGS: Selective Text Refinement in Gaussian Splatting | *WACV 2026* — *Project Page*
Abhinav Raundhal*, Gaurav Behera*, P. J. Narayanan, Ravi Kiran Sarvadevabhatla, Makarand Tapaswi.*

- Proposed STRinGS, a text-aware selective refinement framework for 3D Gaussian Splatting, improving text reconstruction by 63.6% OCR-CER over prior methods.

Beyond Frontiers: Scene-Anomaly Guided Autonomous Exploration | *IROS 2026* — *Project Page*
*Akash Kumbar, **Abhinav Raundhal**, Madhava Krishna.*

- **Best Paper Award** at Bridging Perspectives in Navigation Workshop at IROS 2026.
- Developed SCAGE, a scene-anomaly guided exploration framework that uses learned geometric priors to target poorly reconstructed areas, achieving ∼90% volumetric coverage & 50% lower Chamfer distance than baselines.

PrinGS: Privacy Preservation in Gaussian Splatting | *ICVGIP 2026* — *Project Page*
Gaurav Behera, **Abhinav Raundhal***, Archisha Panda*, Saumya Balina*, Gargi Shroff*, Ponnurangam Kumaraguru.*

- Proposed a representation-level privacy framework for 3D Gaussian Splatting that anonymizes sensitive faces and text across novel viewpoints, achieving a 74.08% reduction in face similarity and 1.00 OCR-CER.

Experience

Cloudphysician | *Computer Vision Engineer* — *Jun. 2026 – Present*

- Developing multimodal representation learning methods for Intensive Care Units (ICUs), integrating heterogeneous clinical modalities like text and video to learn robust representations for downstream critical-care applications.

CVIT, IIIT Hyderabad | *Undergraduate Researcher (Computer Vision)* — *May 2024 – Jun. 2026*

- Conducted research in document understanding and 3D vision, developing methods for text-line segmentation, handwritten document understanding, and text-aware 3D scene reconstruction.

RRC, IIIT Hyderabad | *Undergraduate Researcher (Robotics & 3D Vision)* — *Jan. 2025 – Jun. 2026*

- Developed perception and exploration algorithms for autonomous navigation, with a focus on active exploration, efficient 3D mapping, and high-fidelity scene reconstruction.

ideaForge | *3D Vision Intern* — *May 2025 – Jul. 2025*

- Designed a hardware-efficient 3D reconstruction pipeline for drone-based computer vision systems.

Joly AI | *Computer Vision Intern* — *Jan. 2024 – Apr. 2024*

- Developed learning-based photo enhancement models that replicate photographers' editing styles.

IIIT Hyderabad | *Teaching Assistant* — *Jul. 2024 – May 2026*

- Teaching Assistant for 4 courses: Computer Vision, Statistical Methods in AI, Computer Graphics, and Probability & Statistics; conducted tutorials, prepared and graded assignments, and mentored students.

Projects

DistLearn: Federated Learning | *Distributed Systems, Privacy-Preserving ML* — Github | *Mar. 2025 - May. 2025*

- Built a federated learning system enabling multiple clients to train locally with data privacy, aggregating model updates using methods like FedSGD, FedAvg, FedAdp with SSL/TLS encryption.

6DoF Pose Estimation using Differentiable Rendering | *Mobile Robotics* — Github | *Oct. 2024 - Dec. 2024*

- Developed a framework for 6-DoF pose estimation from meshes using differentiable rendering techniques, focusing on optimizing pose parameters by minimizing silhouette loss.

Simple Renderer using Ray Tracing | *Computer graphics* — Github | *Jan. 2024 - Feb. 2024*

- Built a simple renderer to render scenes using Bounding Volume Hierarchy, texture mapping, direct lighting with point and directional lights, Monte Carlo integration for area light sampling

Skills and Leadership

Languages: Python, C++, C, Go, SQL | **Libraries:** PyTorch, OpenCV, NumPy | **Tools:** Git, gRPC, Blender, Colmap

Cultural Secretary, IIITH | **Technical Secretary**, Students' Parliament IIITH | **Coordinator**, The Dance Crew IIITH | **Design Head**, Electronics & Robotics Club IIITH